

q1_analysis

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Contents

Import and clean dataset

```
pancreatic_df <- read_csv("data/eb1.csv") |>
  janitor::clean_names() |>
  mutate(sex = case_when(sex == 0 ~ "Male",
                        sex == 1 ~ "Female"),
         sex = fct_relevel(sex, "Male"),
         race = case_when(race == 0 ~ "White",
                        race == 1 ~ "Black",
                        race == 2 ~ "Asian",
                        race == 3 ~ "Hispanic"),
         race = fct_relevel(race, "White"),
         op = as.factor(case_when(op == 0 ~ "Body or tail",
                                op == 1 ~ "Whipple operation")),
         ven_res = as.factor(case_when(ven_res == 0 ~ "No",
                                       ven_res == 1 ~ "Yes")),
         neo_chemo = as.factor(case_when(neo_chemo == 0 ~ "No",
                                       neo_chemo == 1 ~ "Yes")),
         neo_xrt = as.factor(case_when(neo_xrt == 0 ~ "No",
                                       neo_xrt == 1 ~ "Yes"))

## Rows: 356 Columns: 13
## -- Column specification -----
## Delimiter: ","
## dbl (13): patid, age, sex, race, liver_test, op, ebl, ven_res, neo_chemo, ne...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(pancreatic_df)
```

```
## # A tibble: 6 x 13
##   patid  age sex    race    liver_test op      ebl ven_res neo_chemo neo_xrt
##   <dbl> <dbl> <fct> <fct>    <dbl> <fct>   <dbl> <fct>   <fct>   <fct>
## 1     1  48.8 Male  White      2.2 Body o~   660 No     No      No
## 2     2   61  Male  White      8.5 Whipl~   800 Yes   Yes     No
```

```
## 3      3  72.3 Female Hispanic      3.5 Body o~  590 No      No      No
## 4      4  58.2 Female White        2.7 Body o~  530 No      No      No
## 5      5  74.8 Male   <NA>         8.9 Body o~  615 No      No      No
## 6      6  36.7 Female Hispanic     3.2 Body o~  680 No      No      No
## # i 3 more variables: les_size <dbl>, or_time <dbl>, prot_energy <dbl>
```

table 1: demographic characteristics + others

```
table1 <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = sex,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",      # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      age ~ "Age (years)",
      race ~ "Race",
      liver_test ~ "Bilirubin Level (in mg/dL)",
      op ~ "Type of Operation",
      ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
      neo_chemo ~ "Chemotherapy Prior to Surgery",
      neo_xrt ~ "Radiation Therapy Prior to Surgery",
      les_size ~ "Tumor Size (in cm)",
      or_time ~ "Operation Time (in mins)"
    )
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Sex**")
table1v2 <- table1 %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",   # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
```

```
style = gt::cell_borders(),
locations = gt::cells_body(columns = everything())) |>
gt::gtsave(filename = "figures_tables/table1.pdf")
```

```
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpP27Xoz/file51c2748c83a9.html screenshot
```

```
system("pdftocrop figures_tables/table1.pdf figures_tables/table1.pdf")
```

```
table1 |>
  as_kable()
```

Table 1: Baseline Characteristics by Sex

Characteristic	Overall N = 356	Male N = 157	Female N = 199	p-value
Age (years)	63 (15)	66 (14)	61 (16)	0.011
Race				<0.001
White	217 (73%)	120 (90%)	97 (59%)	
Asian	9 (3.0%)	5 (3.8%)	4 (2.4%)	
Black	22 (7.4%)	3 (2.3%)	19 (12%)	
Hispanic	49 (16%)	5 (3.8%)	44 (27%)	
Unknown	59	24	35	
Bilirubin Level (in mg/dL)	5.3 (3.6)	5.8 (3.9)	4.8 (3.3)	0.020
Type of Operation				0.035
Body or tail	172 (48%)	66 (42%)	106 (53%)	
Whipple operation	184 (52%)	91 (58%)	93 (47%)	
Estimated Intraoperative Blood Loss (in mL)	658 (92)	675 (89)	644 (93)	0.002
Resection + Reconstruction of Major Vasculature	79 (22%)	47 (30%)	32 (16%)	0.002
Chemotherapy Prior to Surgery	67 (19%)	32 (20%)	35 (18%)	0.5
Radiation Therapy Prior to Surgery	39 (11%)	20 (13%)	19 (9.5%)	0.3
Tumor Size (in cm)	3.28 (2.35)	2.86 (1.71)	3.60 (2.71)	0.043
Operation Time (in mins)	374 (134)	404 (140)	351 (125)	<0.001

```
table1byrace <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = race,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",      # drop min/max to save space
      all_categorical() ~ "{n} ({p})"
    ),
    label = list(
      age ~ "Age (years)",
      sex ~ "Sex",
      liver_test ~ "Bilirubin Level (in mg/dL)",
      op ~ "Type of Operation",
      ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
```

```

neo_chemo ~ "Chemotherapy Prior to Surgery",
neo_xrt ~ "Radiation Therapy Prior to Surgery",
les_size ~ "Tumor Size (in cm)",
or_time ~ "Operation Time (in mins)"
)
) %>%
add_p() %>%
add_overall() %>%
bold_labels() %>%
modify_caption("**Baseline Characteristics by Sex**")

```

59 missing rows in the "race" column have been removed.

```

table1bv2 <- table1byrace %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()) |>
  gt::gtsave(filename = "figures_tables/table1b.pdf")

```

file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmpP27Xoz/file51c26f5787a0.html screenshot

```

system("pdfcrop figures_tables/table1b.pdf figures_tables/table1b.pdf")

table1byrace |>
  as_kable()

```

Table 2: Baseline Characteristics by Sex

Characteristic	Overall N = 297	White N = 217	Asian N = 9	Black N = 22	Hispanic N = 49	p- value
Age (years)	63 (15)	64 (15)	63 (9)	61 (18)	63 (16)	0.9
Sex						<0.001

Characteristic	Overall N = 297	White N = 217	Asian N = 9	Black N = 22	Hispanic N = 49	p- value
Male	133 (45%)	120 (55%)	5 (56%)	3 (14%)	5 (10%)	
Female	164 (55%)	97 (45%)	4 (44%)	19 (86%)	44 (90%)	
Bilirubin Level (in mg/dL)	5.4 (3.7)	5.8 (3.7)	1.9 (2.1)	5.6 (4.8)	4.0 (2.7)	<0.001
Type of Operation						0.4
Body or tail	144 (48%)	109 (50%)	2 (22%)	11 (50%)	22 (45%)	
Whipple operation	153 (52%)	108 (50%)	7 (78%)	11 (50%)	27 (55%)	
Estimated Intraoperative Blood Loss (in mL)	655 (94)	657 (100)	658 (106)	680 (92)	638 (51)	0.2
Resection + Reconstruction of Major Vasculature	61 (21%)	54 (25%)	2 (22%)	4 (18%)	1 (2.0%)	<0.001
Chemotherapy Prior to Surgery	57 (19%)	52 (24%)	0 (0%)	4 (18%)	1 (2.0%)	<0.001
Radiation Therapy Prior to Surgery	30 (10%)	30 (14%)	0 (0%)	0 (0%)	0 (0%)	0.003
Tumor Size (in cm)	3.29 (2.40)	2.98 (1.75)	2.10 (0.74)	5.22 (3.81)	4.01 (3.49)	0.032
Operation Time (in mins)	373 (134)	378 (145)	386 (102)	418 (50)	328 (100)	0.006

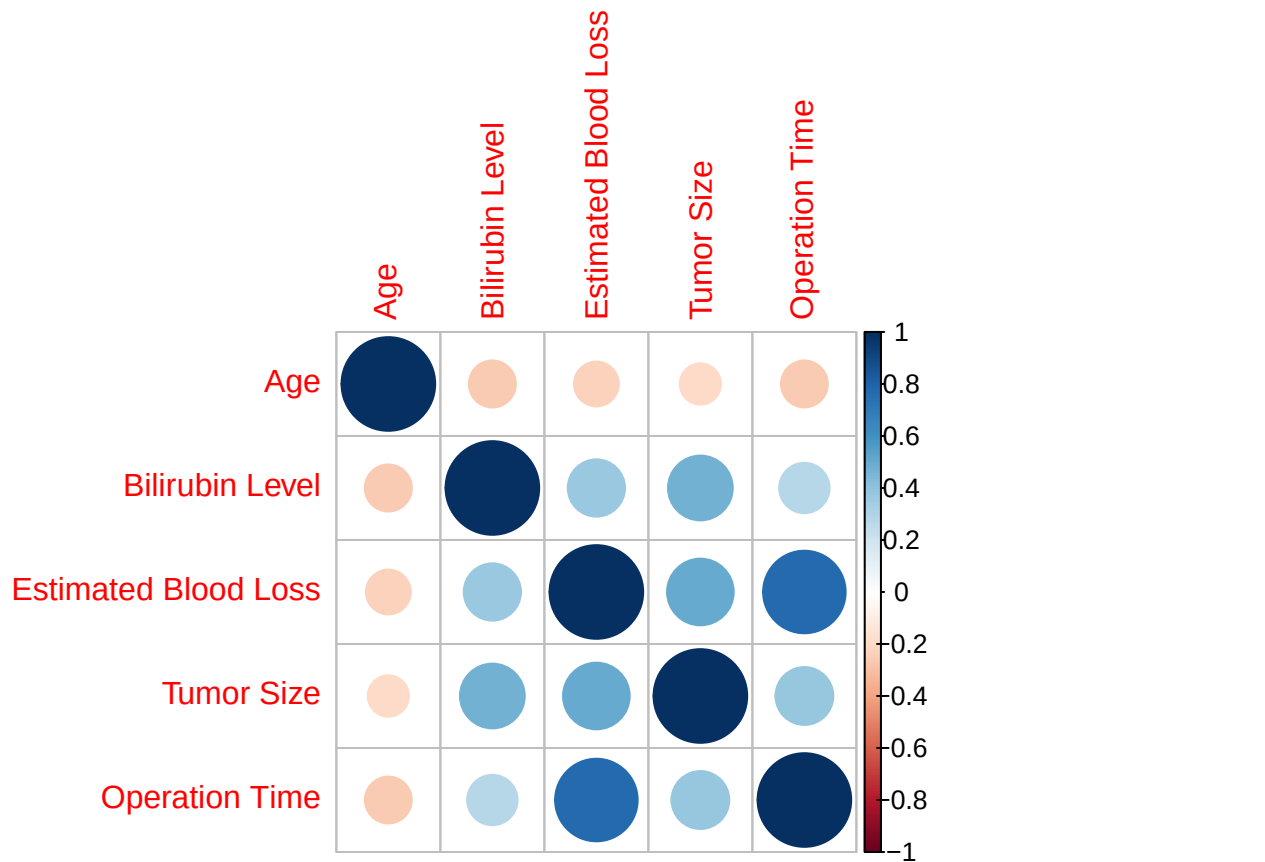
###correlation between predictors and within covariates

Male

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Male") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
pdf("figures_tables/corr_contvars_male.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
## 2
```

```
corrplot(cor_matrix, method = "circle")
```

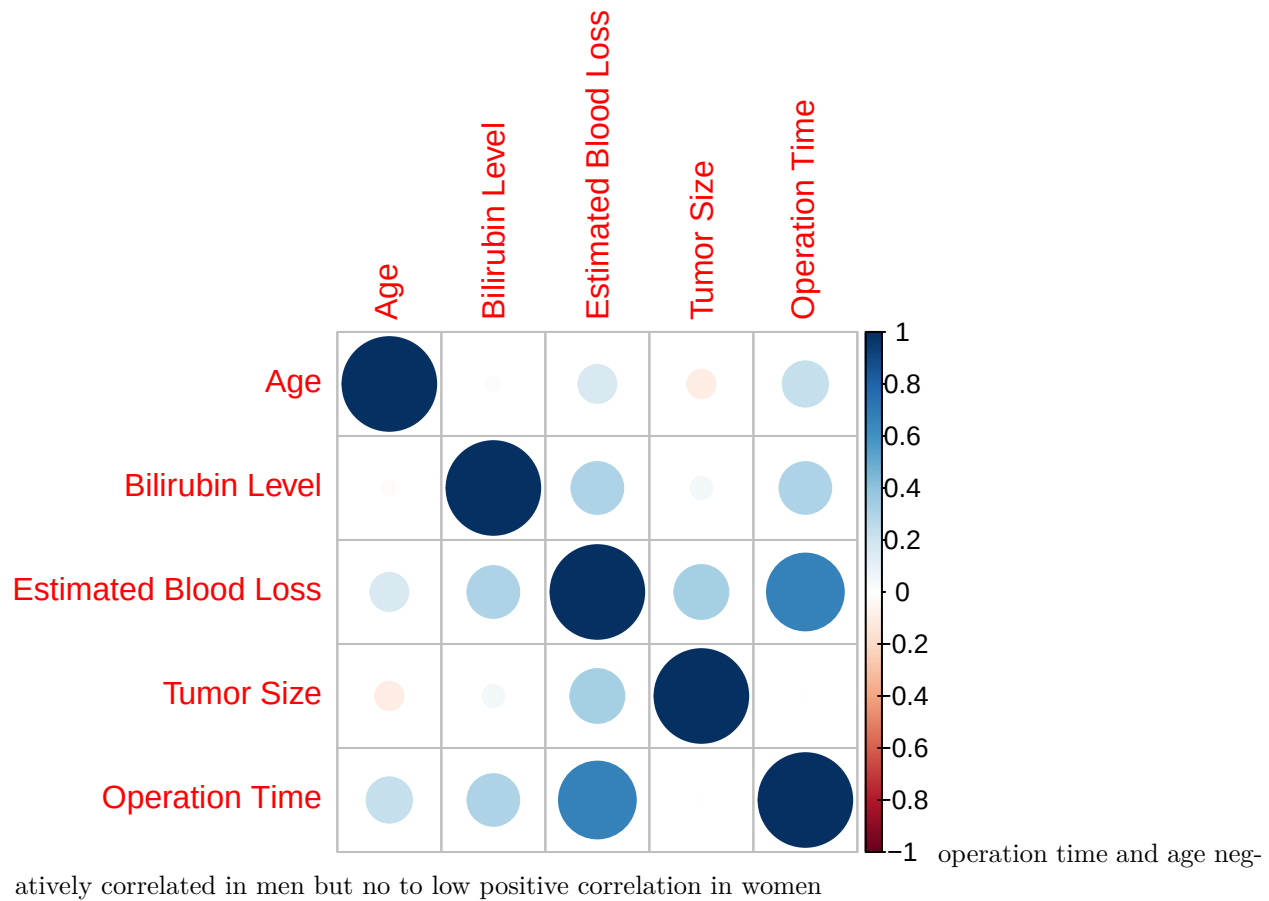


Female

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Female") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
pdf("figures_tables/corr_contvars_female.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
## 2
```

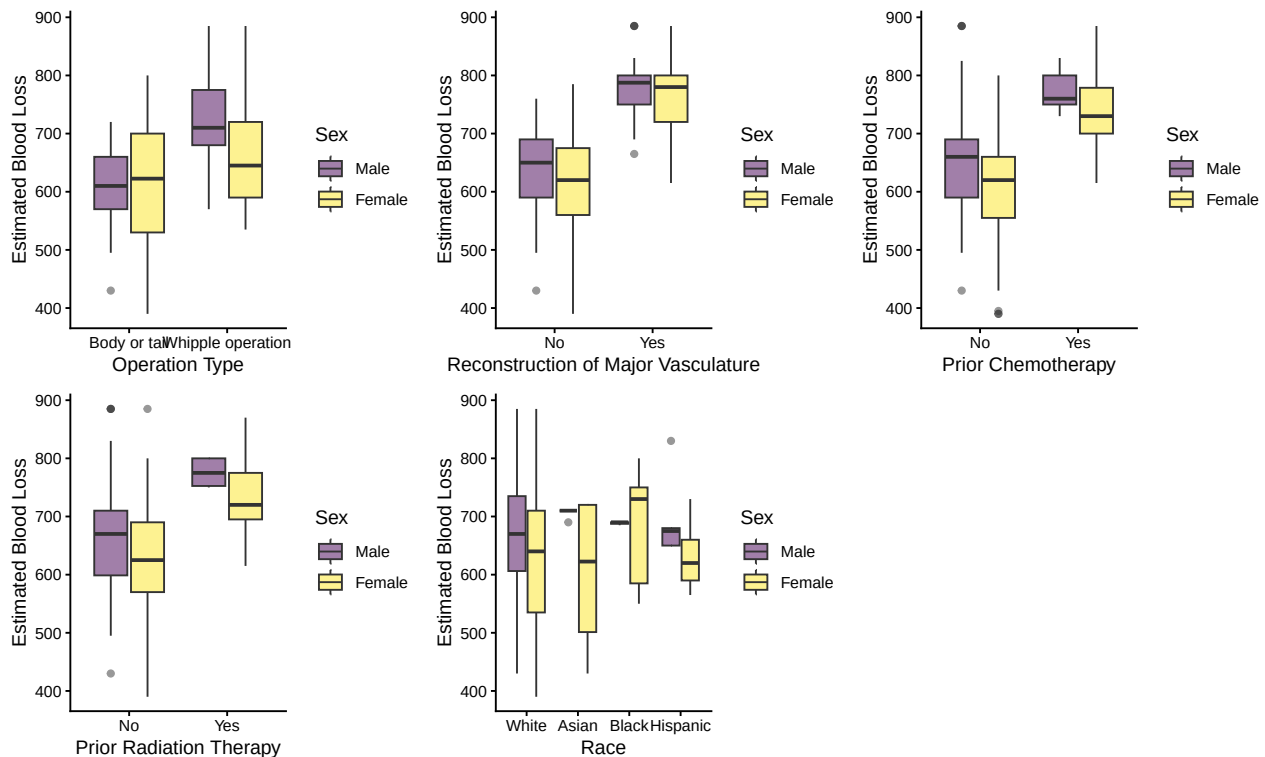
```
corrplot(cor_matrix, method = "circle")
```



EBL Distribution in each categorical covariates

```
colnames <- c("op", "ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior Radiat.
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
    aes(x = get(colnames[x]), y = ebl, fill = sex)) +
    geom_boxplot(alpha = 0.5) +
    labs(x = names[x], y = "Estimated Blood Loss",
      fill = "Sex") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```

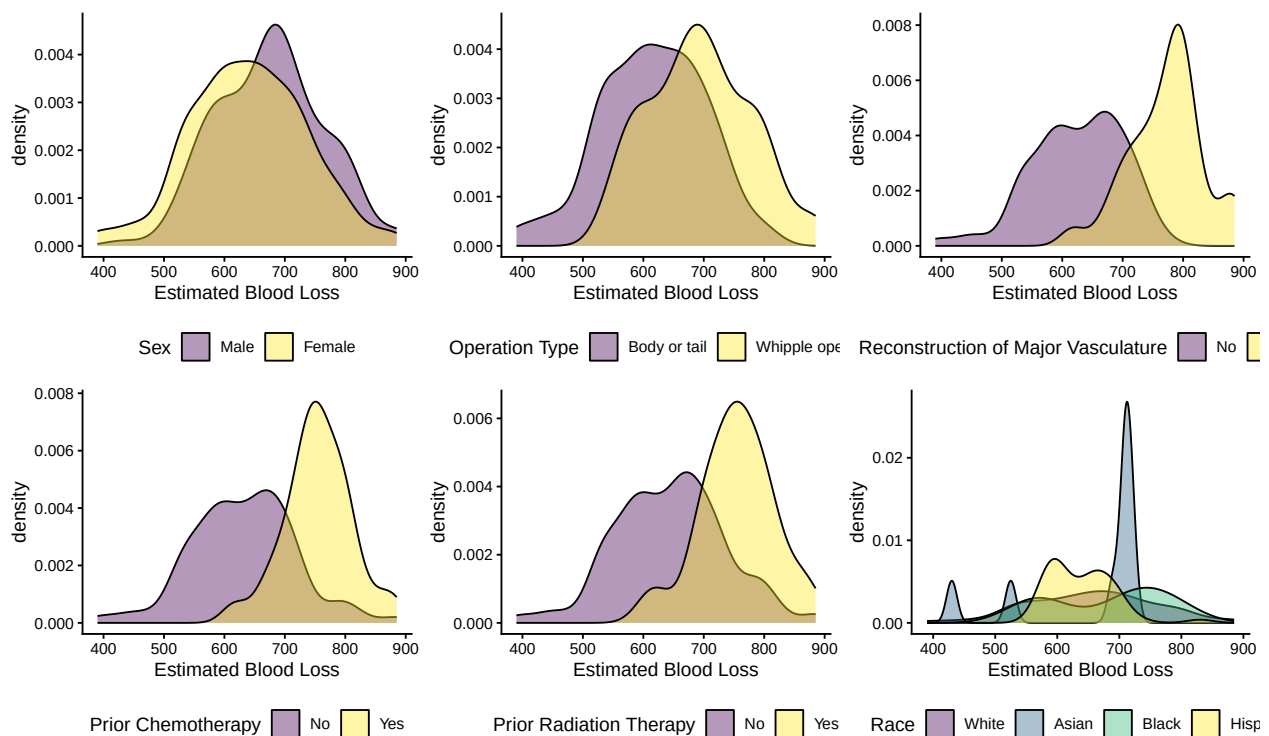


```
pdf("figures_tables/ebl_dist_across_cat_boxplot.pdf",
    width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
## 2
```

```
colnames <- c("sex", "op", "ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                 aes(x = ebl, fill = get(colnames[x]))) +
    geom_density(alpha = 0.4) +
    labs(fill = names[x], x = "Estimated Blood Loss") +
    theme_classic() +
    theme(legend.position = "bottom")
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```



```
pdf("figures_tables/eb1_dist_across_cat_densityplot.pdf",
    width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
## 2
```

EBL lower in female regardless of prior chemotherapy, radiation therapy and major reconstruction status. However, EBL in female vs male are different depending on operation type (effect modification probably, can try interaction term). Low EBL in white and hispanic women. Very few asian and black men recruited in the study so comparison is difficult.

EBL's distribution is relatively normal across each group.

Modeling:

```
lm_v0 <- lm(ebl ~ race*sex + age + sex + op + ven_res + neo_chemo +
            neo_xrt + liver_test*les_size + age*or_time,
            data = pancreatic_df)

summary(lm_v0)

##
## Call:
## lm(formula = ebl ~ race * sex + age + sex + op + ven_res + neo_chemo +
##     neo_xrt + liver_test * les_size + age * or_time, data = pancreatic_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -193.136 -30.879 -1.239 33.745 170.497
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    356.885326   45.012326   7.929 0.0000000000000536 ***
## raceAsian      64.610020   24.967360   2.588   0.01017 *
## raceBlack      54.810438   31.662004   1.731   0.08454 .
## raceHispanic   -22.857469   24.964750  -0.916   0.36067
## sexFemale     -15.566435    7.658829  -2.032   0.04305 *
## age            2.237837    0.705648   3.171   0.00169 **
## opWhipple operation -1.662874    8.894786  -0.187   0.85184
## ven_resYes     25.505856   12.643261   2.017   0.04462 *
## neo_chemoYes    62.184653   12.332633   5.042 0.0000008291816000 ***
## neo_xrtYes     -48.163214   15.769250  -3.054   0.00247 **
## liver_test     -3.120191    1.852476  -1.684   0.09324 .
## les_size        6.833878    2.085215   3.277   0.00118 **
## or_time         0.678997    0.118093   5.750 0.0000000234348958 ***
## raceAsian:sexFemale -107.520328  37.218323  -2.889   0.00417 **
## raceBlack:sexFemale -90.444161   35.137633  -2.574   0.01057 *
## raceHispanic:sexFemale 40.663133   26.807753   1.517   0.13044
## liver_test:les_size  1.106972    0.358578   3.087   0.00222 **
## age:or_time     -0.005360    0.001814  -2.955   0.00339 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 52.95 on 279 degrees of freedom
## (59 observations deleted due to missingness)
## Multiple R-squared:  0.6993, Adjusted R-squared:  0.681
## F-statistic: 38.17 on 17 and 279 DF, p-value: < 0.00000000000000022
```

Type of operation not really significant

```
lm_v1 <- lm(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
            neo_xrt + liver_test*les_size + age*or_time,
            data = pancreatic_df)
anova(lm_v1, lm_v0)
```

```
## Analysis of Variance Table
##
## Model 1: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##       liver_test * les_size + age * or_time
## Model 2: ebl ~ race * sex + age + sex + op + ven_res + neo_chemo + neo_xrt +
##       liver_test * les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      280 782376
## 2      279 782278  1    97.995 0.035 0.8518
```

the model with operation time doesn't provide better fit. So removing that

```
lm_v2 <- lm(ebl ~ race + sex + age + sex + ven_res + neo_chemo +
            neo_xrt + liver_test*les_size + age*or_time,
            data = pancreatic_df)
anova(lm_v2, lm_v1)
```

```
## Analysis of Variance Table
##
## Model 1: ebl ~ race + sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test * les_size + age * or_time
## Model 2: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test * les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      283 831412
## 2      280 782376  3      49037 5.8498 0.0006897 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lm_v3 <- lm(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
            neo_xrt + liver_test*les_size + age + or_time,
            data = pancreatic_df)
anova(lm_v3, lm_v1)
```

```
## Analysis of Variance Table
##
## Model 1: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test * les_size + age + or_time
## Model 2: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test * les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      281 806772
## 2      280 782376  1      24397 8.7313 0.003394 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lm_v4 <- lm(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
            neo_xrt + liver_test + les_size + age*or_time,
            data = pancreatic_df)
anova(lm_v4, lm_v1)
```

```
## Analysis of Variance Table
##
## Model 1: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test + les_size + age * or_time
## Model 2: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##     liver_test * les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      281 808999
## 2      280 782376  1      26624 9.5283 0.002226 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

One with interactions for raceXsex, liver_testXles_size, ageXor_time significantly better
 ###Look at missingness

```
fig2b <- vis_miss(pancreatic_df |> dplyr::select(-prot_energy)) +
  theme(axis.text.x = element_text(vjust = 0))
ggsave("figures_tables/missingness_info.png", fig2b, height = 7, width = 10)
```

Missingness by gender and age

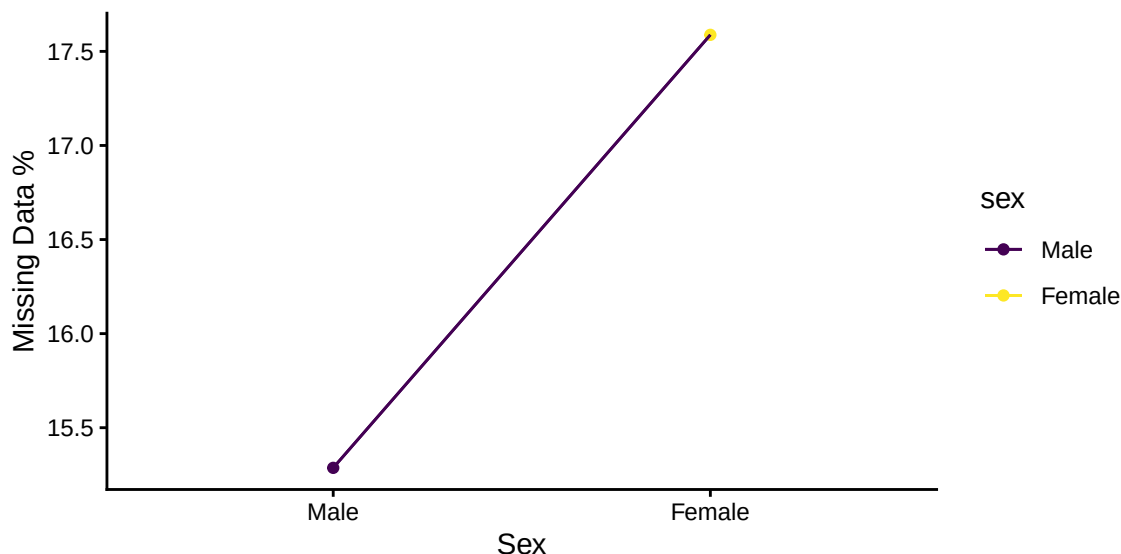
```

race_gender_missing <- pancreatic_df |>
  dplyr::select(-prot_energy) |>
  group_by(sex) |>
  mutate(total = length(unique(patid))) |>
  ungroup() |>
  drop_na(race) |>
  group_by(sex) |>
  summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
  mutate(dummy_group = 1)

missing_by_gender <- ggplot(aes(x = sex, y = missing_pct, group = dummy_group, color = sex),
  data = race_gender_missing) +
  geom_point() +
  geom_line() +
  geom_line() +
  labs(x = "Sex", y = "Missing Data %") +
  theme_classic()

ggsave("figures_tables/missingness_byracegender.pdf", missing_by_gender, height = 3, width =5)
system("pdfcrop figures_tables/missingness_byracegender.pdf figures_tables/missingness_byracegender.pdf")
missing_by_gender

```



Look for other possible sources. - check trend of ebl across each continuous variable to see if its even linear or not, if not linear can try gams

Missingness in different quantile of age or other continuous variables

```

colnames <- c("sex", "op", "ven_res", "neo_chemo", "neo_xrt")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
categorical_missing_pct <- lapply(1:length(colnames), function(x){
  race_gender_missing <- pancreatic_df |>
    dplyr::select(-prot_energy) |>
    group_by(get(colnames[x])) |>
    mutate(total = length(unique(patid))) |>

```

```

ungroup() |>
drop_na(race) |>
group_by(get(colnames[x])) |>
summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
mutate(dummy_group = 1)
colnames(race_gender_missing) <- c(colnames[x], "missing_pct", "dummy_group")
print(head(race_gender_missing))

missing_by_cat <- ggplot(aes(x = get(colnames[x]), y = missing_pct, group = dummy_group),
  data = race_gender_missing) +
  geom_point() +
  geom_line() +
  geom_line() +
  labs(x = names[x], y = "Missing Data %") +
  theme_classic()

return(missing_by_cat)
})

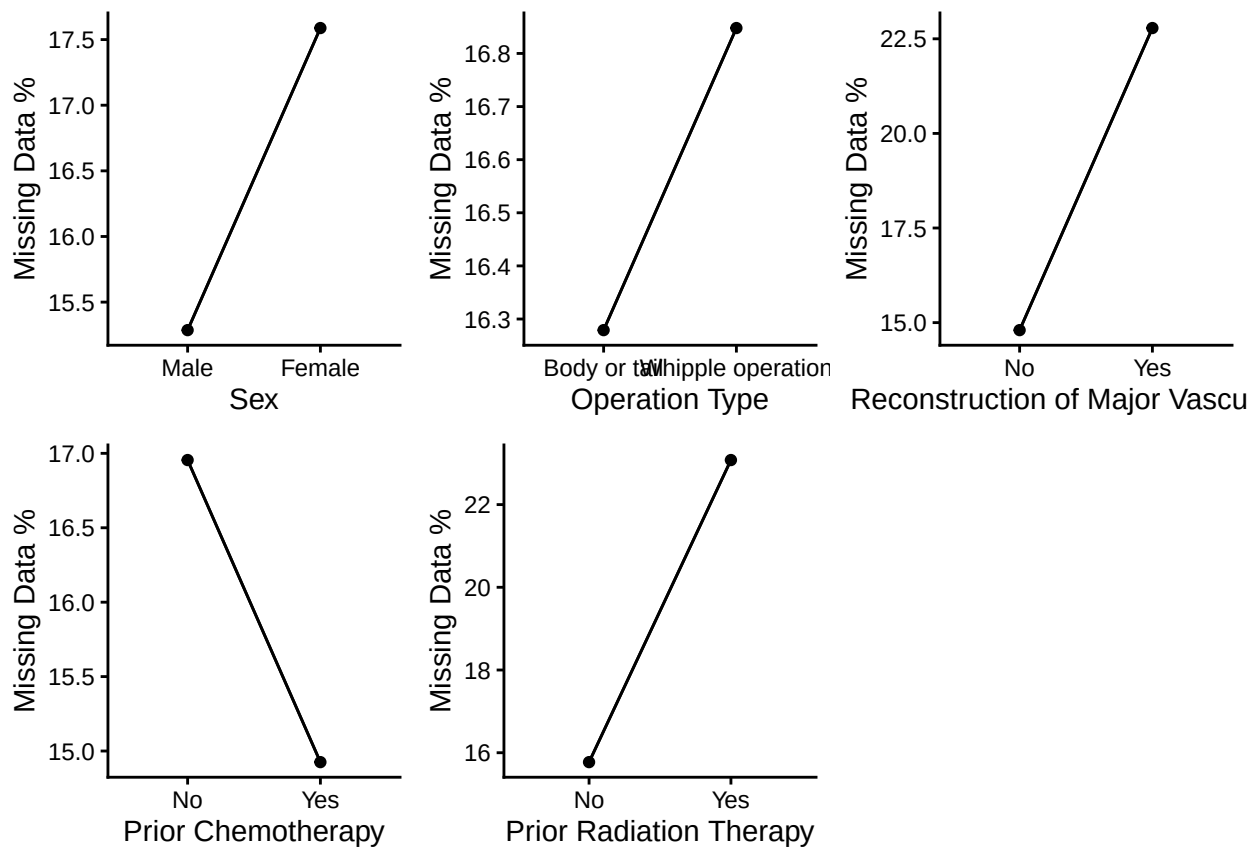
```

```

## # A tibble: 2 x 3
##   sex      missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 Male        15.3        1
## 2 Female      17.6        1
## # A tibble: 2 x 3
##   op                missing_pct dummy_group
##   <fct>                <dbl>      <dbl>
## 1 Body or tail        16.3        1
## 2 Whipple operation   16.8        1
## # A tibble: 2 x 3
##   ven_res missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 No        14.8        1
## 2 Yes       22.8        1
## # A tibble: 2 x 3
##   neo_chemo missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 No        17.0        1
## 2 Yes       14.9        1
## # A tibble: 2 x 3
##   neo_xrt missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 No        15.8        1
## 2 Yes       23.1        1

```

```
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
```



```
pdf("figures_tables/missingness_percategorical_group.pdf",
    width = 9, height = 8)
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
dev.off()
```

```
## pdf
## 2
```

```
colnames <- c("ebl", "age", "liver_test", "les_size", "or_time")
names <- c("Estimated Blood Loss", "Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation
get_quintile_missing_pct <- lapply(1:length(colnames), function(x){
  df_filter <- pancreatic_df %>%
    group_by(sex) |>
    mutate(xcol = get(colnames[x]),
           x_percentile = ntile(xcol, 100)) |>
    group_by(x_percentile, sex) |>
    summarise(n = n(),
              missing_y = sum(is.na(race)),
              group = paste(as.character(min(xcol)), as.character(max(xcol)), sep = "-"),
              covariate = colnames[x],
              pct_missing_y = mean(is.na(race)) * 100)

  plot <- ggplot(df_filter,
                 aes(x = x_percentile, y = pct_missing_y, color = sex)) +
    geom_point(size = 0.5, alpha = 0.5) +
    labs (x = paste(names[x], "(Percentile)" ), y = "% of Missing Race Info",
```

```

    color = "Sex") +
  theme_classic()

  return(plot)
})

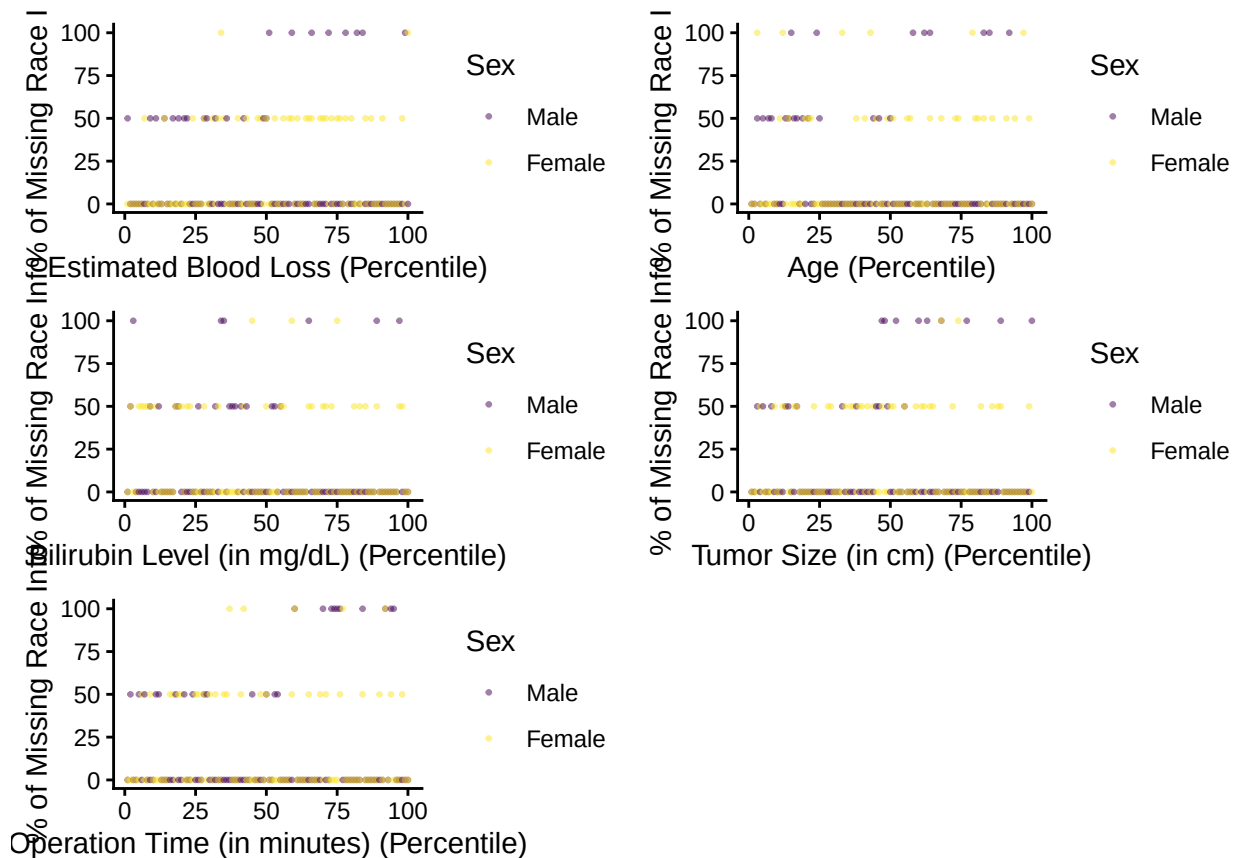
```

```

## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by x_percentile and sex.
## i Output is grouped by x_percentile.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(x_percentile, sex))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

```

```
do.call(gridExtra::grid.arrange, c(get_quintile_missing_pct, ncol = 2))
```



```

pdf("figures_tables/missingness_percontinuous_group.pdf",
    width = 10, height = 8)
do.call(gridExtra::grid.arrange, c(get_quintile_missing_pct, ncol = 2))
dev.off()

```

```

## pdf
## 2

```

The distribution is missing percent against all the continous variables are randomly distributed.

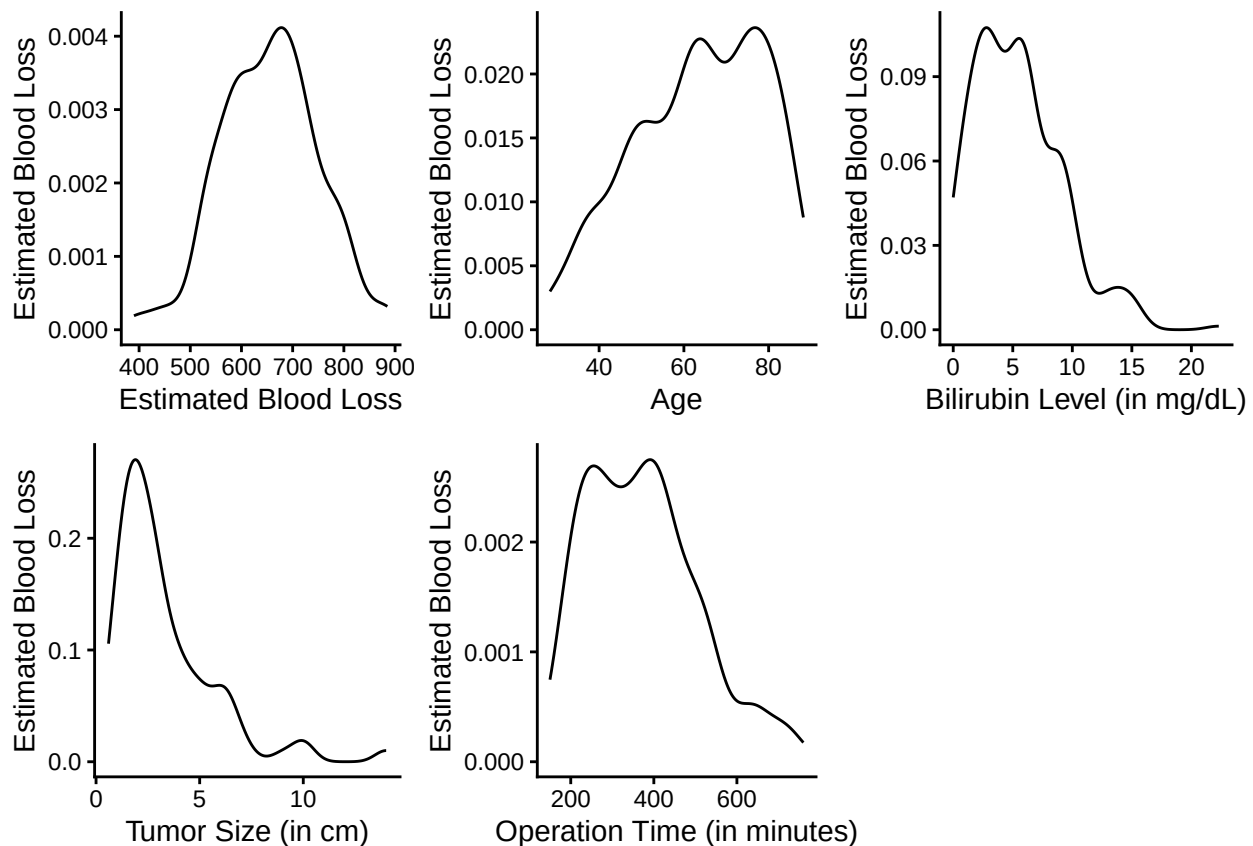
```
mcar_test(pancreatic_df |> dplyr::select(-prot_energy))
```

```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##     <dbl> <dbl>   <dbl>         <int>
## 1     18.9    11  0.0621             2
```

NS → MCAR holds

Tumor size (heavily left-skewed)

```
colnames <- c("ebl", "age", "liver_test", "les_size", "or_time")
names <- c("Estimated Blood Loss", "Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation Time (in minutes)")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
    aes(x = get(colnames[x]))) +
    geom_density(alpha = 0.5) +
    labs(x = names[x], y = "Estimated Blood Loss") +
    theme_classic()
  return(plot)
})
do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```



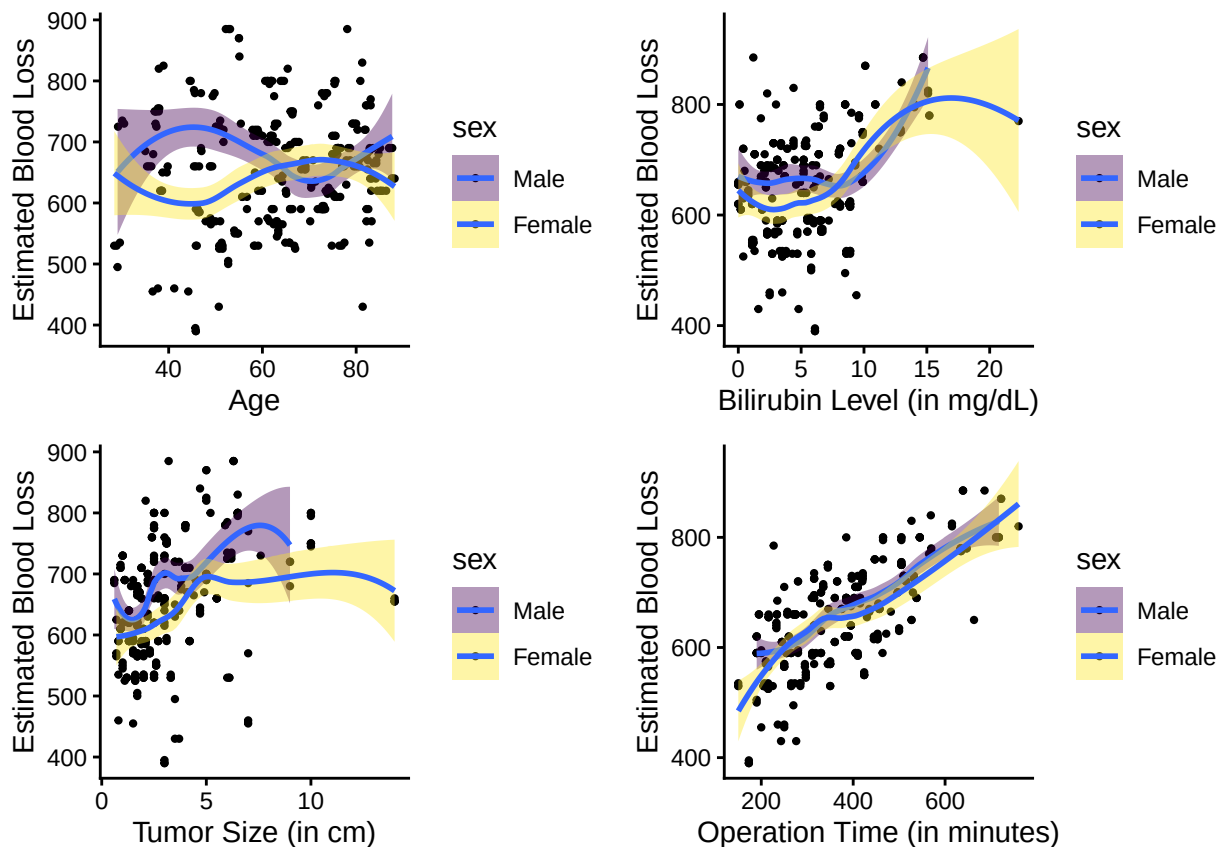
```
pdf("figures_tables/cont_variables_distribution.pdf",
    width = 8, height = 6)
do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
dev.off()

## pdf
## 2

colnames <- c("age", "liver_test", "les_size", "or_time")
names <- c("Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation Time (in minutes)")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
    aes(x = get(colnames[x]), y = ebl, fill = sex)) +
    geom_point(size = 0.8) +
    geom_smooth() +
    labs(x = names[x], y = "Estimated Blood Loss") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 2))

## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



```
pdf("figures_tables/ebl_versus_cont_variable.pdf",
    width = 8, height = 6)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
dev.off()
```

```
## pdf
## 2
```

Bilirubin level - minimal association at smaller amount, increased association after 5

GAM

```
gam_v1 <- gam(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
              neo_xrt + s(liver_test) + s(les_size) + ti(liver_test, les_size) + age*or_time,
              data = pancreatic_df)
```

```
summary(gam_v1)
```

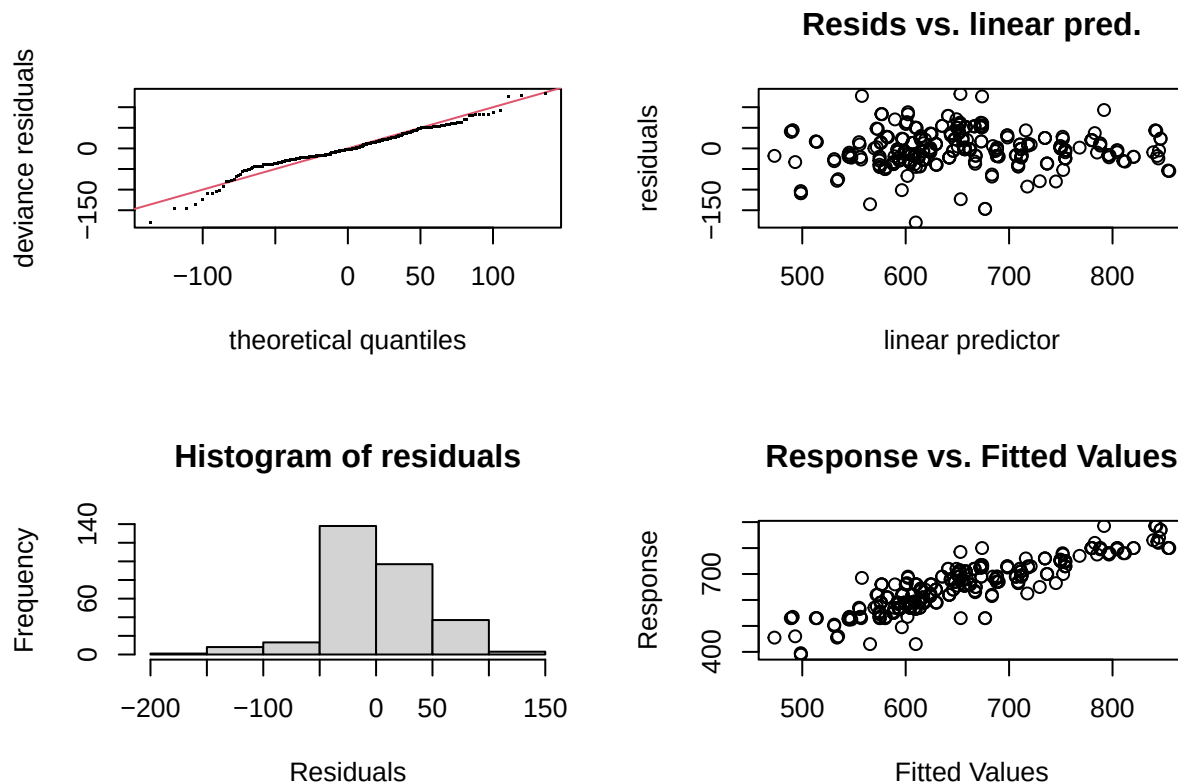
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##       s(liver_test) + s(les_size) + ti(liver_test, les_size) +
##       age * or_time
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   385.328962  46.610102  8.267 0.00000000000000731 ***
## raceAsian      59.183352  23.061068  2.566  0.01084 *
```

	Estimate	Std. Error	t value	Pr(> t)
## raceBlack	92.320040	33.600878	2.748	0.00643 **
## raceHispanic	-28.251639	23.158851	-1.220	0.22361
## sexFemale	-39.053208	7.810796	-5.000	0.00000106047846007 ***
## age	2.350020	0.732266	3.209	0.00150 **
## ven_resYes	23.358724	12.908975	1.809	0.07154 .
## neo_chemoYes	91.958297	13.448298	6.838	0.000000000005791876 ***
## neo_xrtYes	-54.898323	16.794557	-3.269	0.00123 **
## or_time	0.673943	0.128523	5.244	0.00000032786157660 ***
## raceAsian:sexFemale	-77.194775	34.362407	-2.246	0.02552 *
## raceBlack:sexFemale	-118.205830	37.756804	-3.131	0.00194 **
## raceHispanic:sexFemale	62.862126	25.047774	2.510	0.01270 *
## age:or_time	-0.005493	0.001970	-2.789	0.00569 **

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Approximate significance of smooth terms:
##              edf Ref.df    F      p-value
## s(liver_test)   8.744  8.964  2.924    0.00223 **
## s(les_size)     7.177  8.003 12.643 < 0.0000000000000002 ***
## ti(liver_test,les_size) 9.055 10.507  3.828    0.000037 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.755   Deviance explained = 78.7%
## GCV = 2476.6   Scale est. = 2151.6       n = 297
```

```
gam.check(gam_v1)
```



```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 26 iterations.
## The RMS GCV score gradient at convergence was 0.0008729279 .
## The Hessian was positive definite.
## Model rank =  48 / 48
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##              k'   edf k-index      p-value
## s(liver_test)   9.00 8.74   0.72 <0.0000000000000002 ***
## s(les_size)     9.00 7.18   0.86      0.01 **
## ti(liver_test,les_size) 16.00 9.05   0.71 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
gam_v2 <- gam(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
             neo_xrt + s(liver_test) + s(log(les_size)) + ti(liver_test, log(les_size)) + age*or_time,
             data = pancreatic_df)

summary(gam_v2)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##       s(liver_test) + s(log(les_size)) + ti(liver_test, log(les_size)) +
##       age * or_time
##
## Parametric coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)   384.106663   47.019113   8.169 0.000000000000000153 ***
## raceAsian      61.880561   23.200309   2.667    0.00815 **
## raceBlack     179.094518   64.472094   2.778    0.00588 **
## raceHispanic  -30.810399   22.914283  -1.345    0.17997
## sexFemale     -39.277694    7.790853  -5.042 0.0000008848807599 ***
## age           2.290556    0.741738   3.088    0.00224 **
## ven_resYes    20.786987   13.588060   1.530    0.12733
## neo_chemoYes  86.917785   13.962049   6.225 0.0000000020046328 ***
## neo_xrtYes   -47.759932   16.671364  -2.865    0.00453 **
## or_time       0.668391    0.129041   5.180 0.0000004562566908 ***
## raceAsian:sexFemale -78.996243   34.446482  -2.293    0.02266 *
## raceBlack:sexFemale -204.522859   67.038796  -3.051    0.00253 **
## raceHispanic:sexFemale 66.264401   25.345222   2.614    0.00948 **
## age:or_time   -0.005203    0.001995  -2.609    0.00963 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F        p-value
## s(liver_test)    8.165  8.774  3.474    0.000906 ***
## s(log(les_size))  8.749  8.937 12.870 < 0.0000000000000002 ***
## ti(liver_test,log(les_size)) 14.943 15.423  5.311 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) = 0.765   Deviance explained = 80%
## GCV = 2444.6   Scale est. = 2067.1    n = 297
```

```
###Compare fit
```

```
allmods_diag <- data.frame(
  Model = c("LM", "GAM_v1", "GAM_v2"),
  AIC    = c(AIC(lm_v1), AIC(gam_v1), AIC(gam_v2)),
  BIC    = c(BIC(lm_v1), BIC(gam_v1), BIC(gam_v2)),
  `log(Likelihood)` = c(as.numeric(logLik(lm_v1)), as.numeric(logLik(gam_v1)),
                        as.numeric(logLik(gam_v2))),
```

```

DF      = c(attr(logLik(lm_v1), "df"),
             attr(logLik(gam_v1), "df"),
             attr(logLik(gam_v2), "df"))
) |>
  kable(caption = "",
        booktabs = FALSE, longtable = FALSE, linesep = "", digits = 2, full_width = FALSE) |>
  kable_styling(latex_options = "hold_position", font_size = 12)

#save_kable(allmods_diag, file = "figures_tables/model_comparison.pdf")
#system("pdftocrop figures_tables/model_comparison.pdf figures_tables/model_comparison.pdf")

allmods_diag

```

Table 3:

Model	AIC	BIC	log.Likelihood.	DF
LM	3218.13	3284.62	-1591.06	18.00
GAM_v1	3160.18	3307.84	-1540.12	39.98
GAM_v2	3154.03	3327.11	-1530.16	46.86

Based on this, GAM_v1 is chosen (but BIC is higher, so need to look into this)

```

pdf("figures_tables/.pdf", width = 8, height = 8)
plot(gam_v1, shade = TRUE, pages = 1,
     xlab = "liver_test", ylab = "s(liver_test), edf = 8.735")
plot(gam_v1, shade = TRUE, pages = 1,
     xlab = "les_size", ylab = "s(les_size), edf = 7.134")
dev.off()

```

```

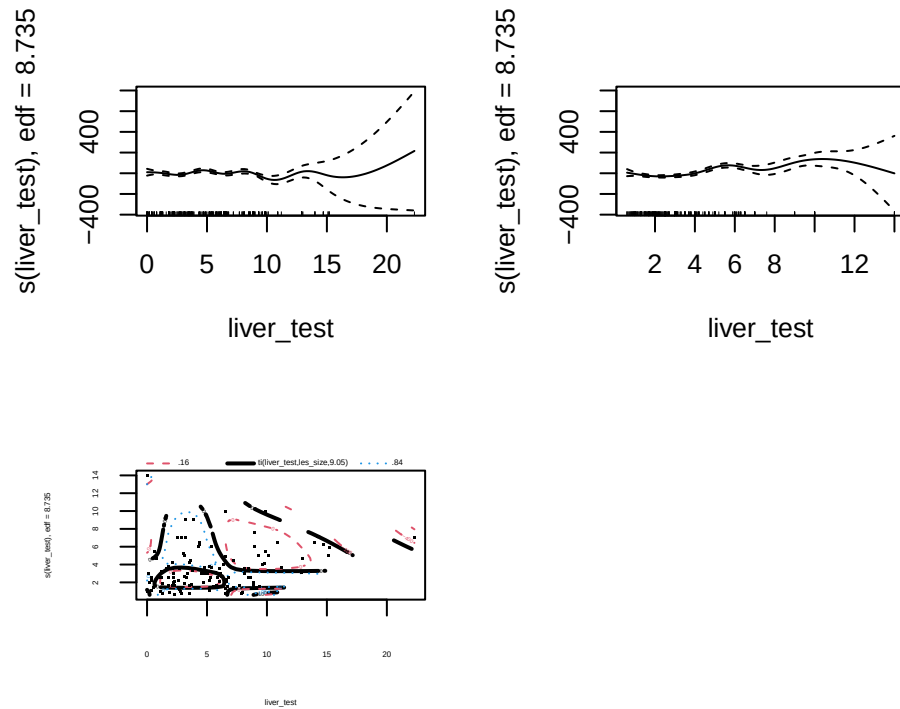
## pdf
## 2

```

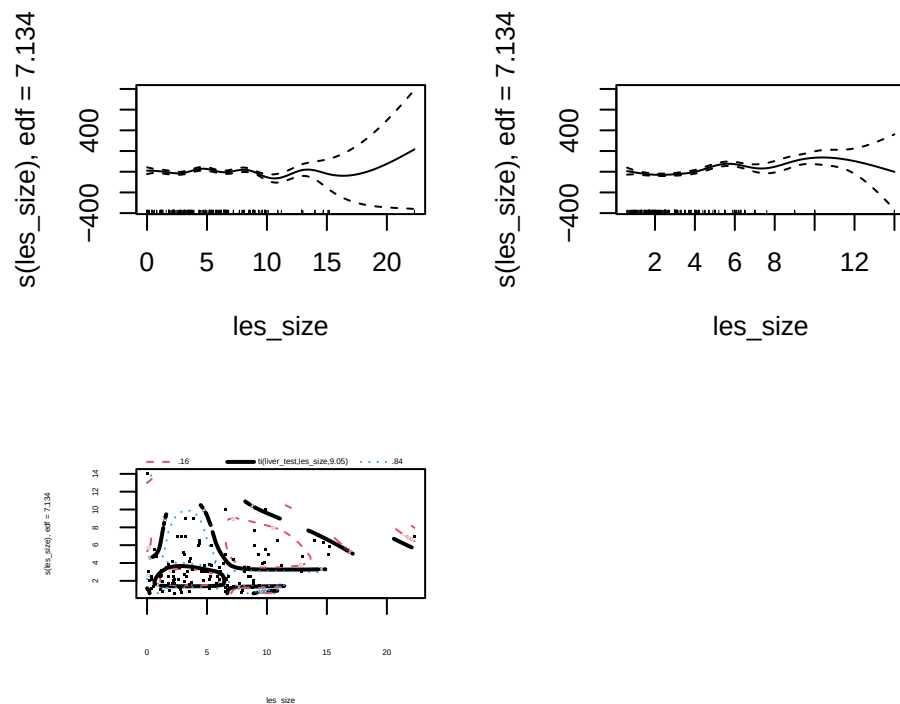
```

plot(gam_v1, shade = TRUE, pages = 1,
     xlab = "liver_test", ylab = "s(liver_test), edf = 8.735")

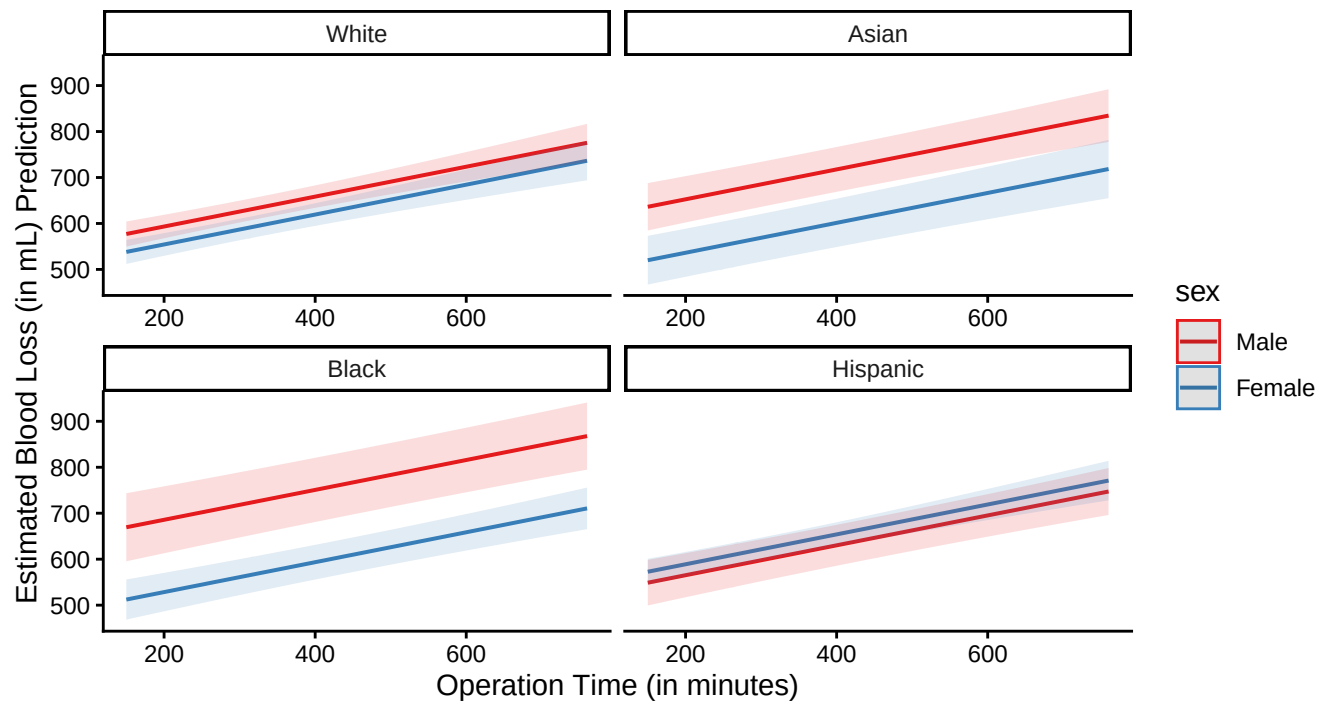
```



```
plot(gam_v1, shade = TRUE, pages = 1,
     xlab = "les_size", ylab = "s(les_size), edf = 7.134")
```

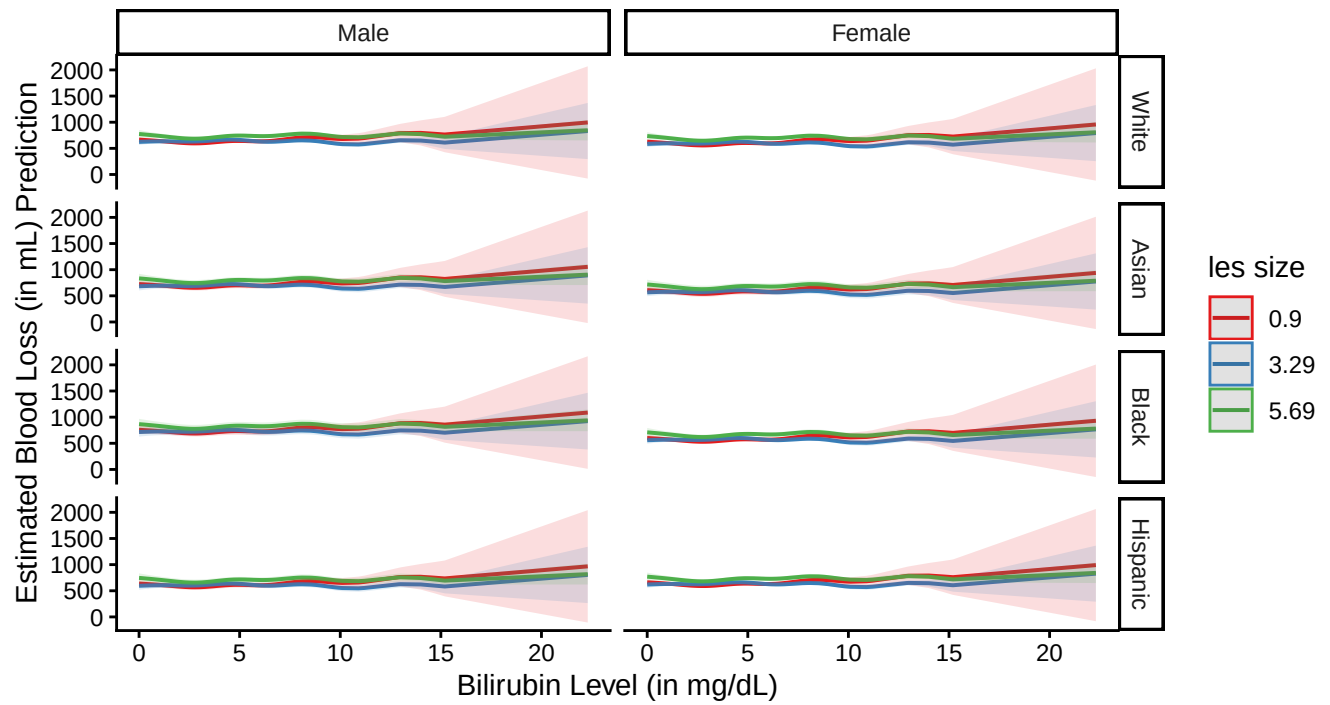


```
plot_model(gam_v1, type = "pred", terms = c("or_time", "sex", "race")) +
  labs(title = "",
       y = "Estimated Blood Loss (in mL) Prediction",
       x = "Operation Time (in minutes)") +
  theme_classic()
```



```
ggsave("figures_tables/eb1_pred_race_gender_operatn_time.pdf", height = 4, width = 8)
```

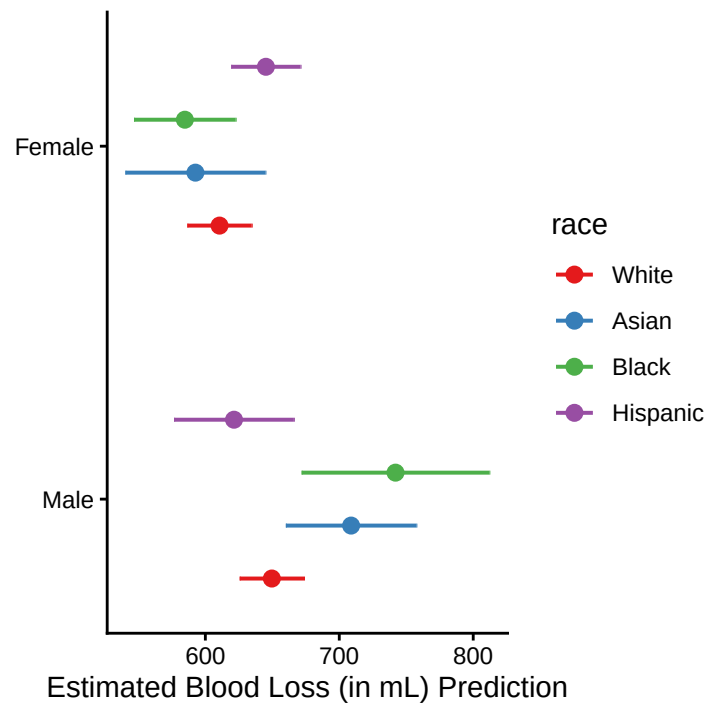
```
plot_model(gam_v1, type = "pred", terms = c("liver_test", "les_size", "sex", "race")) +
  labs(title = "",
        y = "Estimated Blood Loss (in mL) Prediction",
        x = "Bilirubin Level (in mg/dL)") +
  theme_classic()
```



```
ggsave("figures_tables/eb1_pred_race_gender_liver_test_tumorsize.pdf", height = 7, width = 8)
```

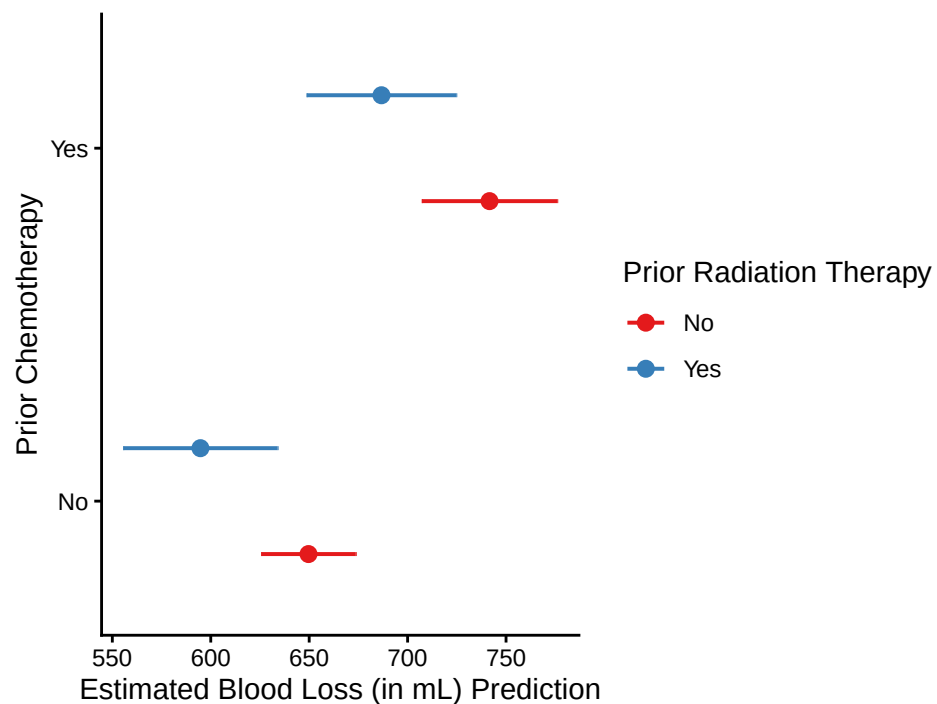
Same trend in men and women across all 4 races

```
plot_model(gam_v1, type = "pred", terms = c("sex", "race"), dodge = 0.6) +
  labs(title = "",
        y = "Estimated Blood Loss (in mL) Prediction",
        x = "") +
  theme_classic() +
  coord_flip()
```



```
ggsave("figures_tables/eb1_pred_race_gender.pdf", height = 3, width = 4)
```

```
plot_model(gam_v1, type = "pred", terms = c("neo_chemo", "neo_xrt"), dodge = 0.6) +
  labs(title = "",
        y = "Estimated Blood Loss (in mL) Prediction",
        x = "Prior Chemotherapy", color = "Prior Radiation Therapy") +
  theme_classic() +
  coord_flip()
```



```
ggsave("figures_tables/eb1_pred_chem_rad_therapy.pdf", height = 3, width = 5)
```

Consider that normal bilirubin level is 0.2-1.3. Compare the normal state with degree of increase or decrease from there